Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

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Advanced Algorithms and Parallel Programming 1st mid-term test– November 9th 2023

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:15m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

Answer the following questions about parallel application modeling and briefly explain (without an explanation, the answer will be considered invalid)

A. Explain why, in the PRAM model, the number of instructions executed is used as time complexity. (2)

B. Define the cost C when a sequential code is accelerated with p processors. (1)

C. What is it meant with a fixed time model? Is it referring to Gustafson or Amdahl's law? (1)

Exercise n. 2

Answer the following questions about dynamic programming algorithms and briefly explain (without an explanation, the answer will be considered invalid)

A. Why the BDD data structure can be classified as an example of dynamic programming. (2)

B. What kind of spatial complexity has the longest common subsequence problem? (1). **This question can be skipped in case of a passed challenge.**

C. Why does spatial complexity matter for all dynamic programming algorithms? (1)

Exercise n. 3

Answer the following questions about advanced algorithms and briefly explain (without an explanation, the answer will be considered invalid)

A. Please discuss the competitive analysis. For which class of algorithm does it matter? (2)

B. Describe how the search for skip-list data structure works. (2)

Exercise n. 4

Answer the following questions about the algorithm analysis and briefly explain (without an explanation, the answer will be considered invalid)

A. Please explain the following theorem and why does not hold for the counting sort algorithm:
Theorem: "Any decision tree that can sort n elements must have a height of $W(n \log n)$" (2)

B. What is meant by "potential method"? To which class of algorithms could be applied? Examples? (2)